

4(a)	a body/mass/object continues (at rest or) at constant/uniform velocity unless acted on by a resultant force	B1
4(b)(i)	initial momentum = final momentum $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	C1
	$0.60 \times 100 - 0.80 \times 200 = -0.40 \times 100 + v \times 200$ $v = (-) 0.3(0) \text{ m s}^{-1}$	A1
4(b)(ii)	<u>kinetic</u> energy is not conserved/is lost (but) <u>total</u> energy is conserved/constant or some of the (initial) <u>kinetic</u> energy is transformed into other forms of energy	B1

Question	Answer	Marks
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